

CAPM:

Individual stocks returns depend on market risk(Beta) and excess returns (alpha) don't exist.

$$r_{i,t} - r_f = \alpha + \beta(r_{m,t} - r_f) + \epsilon_t$$

Method:

```
import seaborn as sns
import statsmodels.api as sm
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib
```

1.Download SP 500 as the market rate

Download 10 individual stocks from different fields [TSLA, MU, NFLX, AAPL, JPM, COST, NEE, ABBV, LMT]

Use Bond Rate as risk-free rate(rf)

```
raw_data = yf.download(ticker, start=start_date, end=end_date)
```

2.Calculate Log returns

```
rt = np.log(Pt/Pt-1)
log_returns = np.log(price_data / price_data.shift(1))
log_returns = log_returns.replace([np.inf, -np.inf], np.nan).dropna()
log_returns = log_returns.dropna()
```

3.Individual excess return y= rstock-rf, Market excess return: x=rmarket-rf

4.Regression Model

Using Statsmodel.OLS get alpha, beta, R squared, p_alpha, p_beta

PValue<0.05, significant, PValue>0.05, insignificant

H0: alpha=0, excess returns don't exist

for stock in stock_list:

```
stock_excess = log_returns[stock]-rf_daily
market_excess = log_returns[market_rate]-rf_daily
stock_excess, market_excess = stock_excess.align(
    market_excess, join="inner")
```

```
data = pd.DataFrame(
    {"stock": stock_excess, "market": market_excess}).dropna()
data = data.dropna()
```

```
x = data["market"]
x_const = sm.add_constant(x)
y = data["stock"]
model = sm.OLS(y, x_const).fit()
```

```
alpha = model.params["const"]
beta = model.params["market"]
r2 = model.rsquared
p_alpha = model.pvalues["const"]
p_beta = model.pvalues["market"]
```

```
residuals = model.resid
```

5.Rolling Beta

Window Length= 50 days

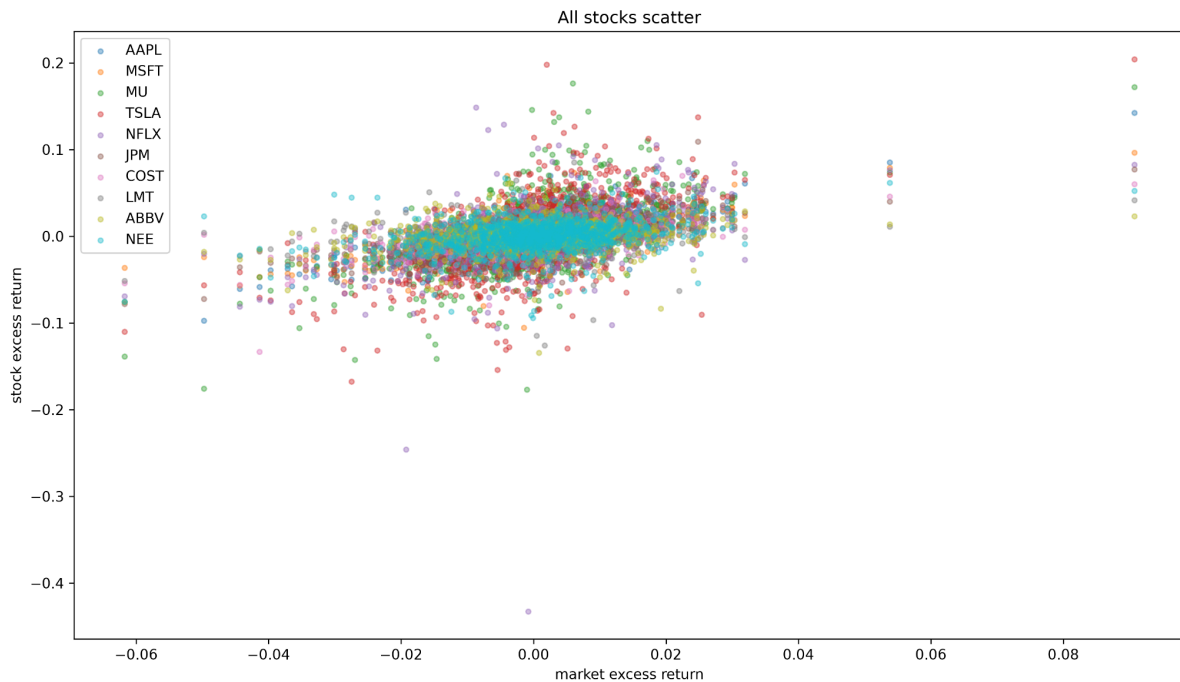
Get a series of Beta with time changes—see whether individual stock risks will change as market changes

```
roll_data = pd.DataFrame(
    {"market": market_excess, "stock": stock_excess}).dropna()
window_length = 50
beta_list = []
for i in range(len(roll_data)):
    if i < window_length-1:
        beta_list.append(np.nan)
    else:
        window_slice = roll_data.iloc[i-window_length+1:i+1]
        window_slice = window_slice.dropna()
        if len(window_slice) == 0 or len(window_slice) < 5:
            beta_list.append(np.nan)
        else:
            x_win = window_slice["market"]
            x_const_win = sm.add_constant(x_win)
            y_win = window_slice["stock"]
            window_model = sm.OLS(y_win, x_const_win).fit()
            beta_list.append(window_model.params["market"])
rolling_series = pd.Series(beta_list, index=roll_data.index)
rolling_beta_df[stock] = rolling_series
```

Results:

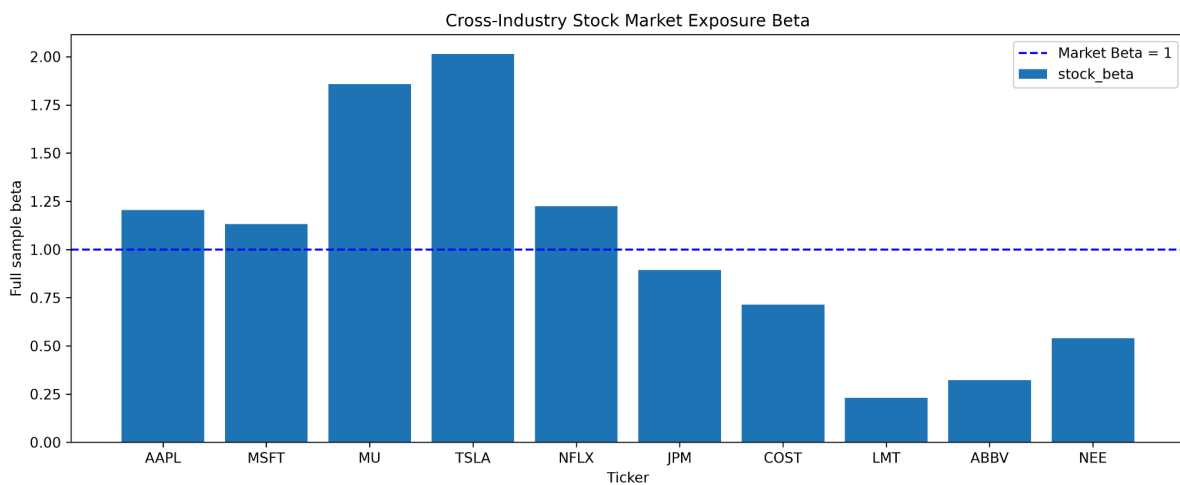
1.CAPM Results

Stock	Alpha	Beta	R2	P_alpha	P_beta
AAPL	0.000118	1.204921	0.549452	0.72177	7.9E-219
MSFT	-0.00019	1.130903	0.50782	0.57114	8.4E-195
MU	0.001468	1.85709	0.339772	0.061387	6.4E-115
TSLA	-0.00023	2.013906	0.337253	0.788138	6.9E-114
NFLX	-0.00027	1.22303	0.215624	0.701848	5.13E-68
JPM	0.000277	0.892853	0.384962	0.418152	3.3E-134
COST	0.000372	0.714139	0.281789	0.280906	5.17E-92
LMT	0.000112	0.229777	0.027981	0.783898	2.57E-09
ABBV	0.00053	0.323174	0.055135	0.190955	3.72E-17
NEE	-7.8E-05	0.540811	0.115986	0.863531	2.13E-35



Beta Analysis:

All $P_{\text{beta}} < 0.05$ indicating individual stock returns significantly get affected by market portfolio.



Specifically,

Beta > 1: AAPL, MSFT, MU, TSLA, NFLX (Aggressive, more volatile than the market)

— Startups, Tech, Growing companies

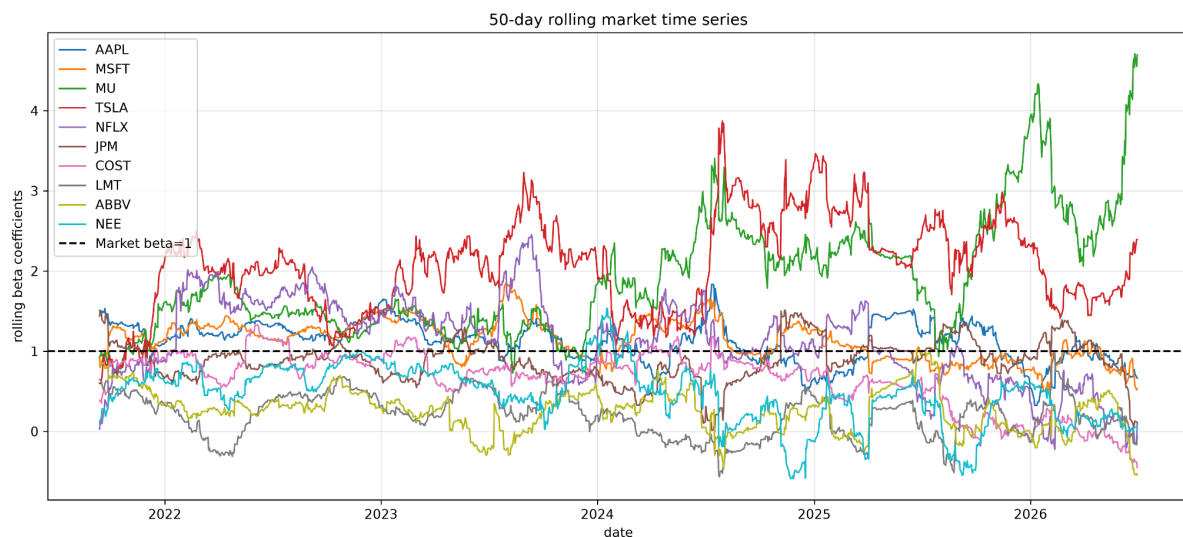
Beta < 1: JPM, COST, LMT, ABBV, NEE (Defensive, less volatile than the market)

— Utility, Necessity, Retail, Medicine, Defense

The combination of different stocks with different beta can reach diversification in order to confront market volatility.

From Big to Small: TSLA, MU, NFLX, AAPL, JPM, COST, NEE, ABBV, LMT

Higher Beta means more sensitive to the market changes. For example, LMT has the lowest Beta, indicating that the market is having a limited effect on the company returns. With the other same conditions, a higher beta means a higher expected return. As TSLA is having a much higher beta than NEE, theoretically investors have a higher expectation on TSLA's long term return.



From 2022-2026, we can see that beta value is not a constant. Every stock's beta is continuously changing. Thus, individual stocks react differently to the market as the market environment, business cycle or company situation changes.

The original CAPM states beta as constant and fixed, the rolling beta chart concludes that the model's ideal hypothesis has difference with the reality.

Sometimes, the individual stock beta value can be higher or lower than 1 as time changes. Therefore, we should introduce dynamic motoring instead of relying solely on past data.

The trends of the individual stocks are overly similar. During crisis or high volatility periods, individual stocks are more likely to follow the market.

Alpha Analysis:

H0: $\alpha=0$

All the individual stocks have higher-than-0.05 P_{α} value. Only MU's P_{α} equals 0.0614, sitting at the edge of 10% significant level. Under the 5% significance level, there is no significant evidence to reject H0. Based on all the sample data, no individual stocks show abnormal excess returns statistically. This aligns with the CAPM hypothesis. The different alpha values, whether negative or positive, are only random volatility instead of better performance than the market.

R Squared Analysis:

Every stock has a really different R squared value. For stocks like AAPL and MSFT that have higher R Squared values, they have a higher linkage with the market. LMT, ABBV, and other companies have lower R squared. meaning the changes of stock price is more explained by the company information.

The analysis only introduces a single factor in to CAPM and doesn't take value, scale, size factors into account. Also, the use of daily returns may have loud noises, affecting the significance of alpha.